

Robust Spectral Recovery for Dynamical Sampling

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Abstract—We study the spectral recovery problem for dynamical sampling on a finite cyclic grid. Given time snapshots obtained from a fixed uniform spatial subsampling of the orbit $x_\ell = A^\ell f$, we aim to recover the spectrum of the unknown circular convolution operator A . However, in the presence of outliers, even in only a few snapshots, existing approaches often struggle to recover the spectrum. We address this challenge by proposing a novel robust spectral recovery model in the presence of time-sparse corruptions. We propose a robust pipeline that lifts the problem to a sequence of robust low-rank Hankel recovery and completion tasks, followed by Prony-type spectral estimation. Numerical experiments confirm the accurate spectral recovery of the proposed approach and exhibit its superior robustness against state-of-the-art under various settings.

Index Terms—Dynamical sampling, Low-rank Hankel matrix, Robust spectral recovery.

I. INTRODUCTION

Spatiotemporal data acquisition problems often face a fundamental constraint: deploying dense sensor arrays is expensive, while collecting repeated measurements in time is comparatively cheap. *Dynamical sampling* formalizes this time-spatial trade-off by studying recovery of the underlying dynamics and/or an unknown initial state from coarse spatial samples collected across multiple time steps [1]–[11]. The paradigm is motivated by applications in sensor networks, imaging and inverse problems [12]–[16], where the observed field evolves according to a known physical mechanism but only a subset of spatial locations can be measured repeatedly [1]. A prototypical example is the heat/diffusion equation, where the evolution acts as a strong low-pass filter over time, enabling reconstruction from fewer spatial sensors when temporal oversampling is available [15].

A central and practically relevant class of dynamics is spatially invariant evolution [17], [18], where the evolution operator is a convolution. Convolution models arise naturally from discretizations of translation-invariant PDEs (e.g., diffusion), from linear time-invariant systems, and from shift-invariant blurs in signal and image processing. In this setting,

the driving operator A is completely characterized by a short “filter” a , and the eigenvectors are the Fourier modes; hence A is diagonalized by the discrete Fourier transform (DFT), with eigenvalues given by $\hat{a}$. This structure makes the model both expressive and computationally attractive: applying A can be done rapidly via FFTs, and the unknown dynamics can be interpreted through its spectrum $\hat{a}$.

Within dynamical sampling, convolution operators have received special attention because they allow *system identification* from subsampled spatiotemporal measurements. In particular, [19], [20] study identification of an unknown convolution filter a and initial state f from samples of the evolving states $f, Af, \dots, A^{L-1}f$ taken on a sublattice, and establish a Prony-type spectral method with perturbation analysis. More broadly, the modern theory provides conditions under which the driving operator and/or the initial state can be recovered from time-spatial samples, clarifying when temporal oversampling compensates for spatial undersampling [2].

While existing dynamical sampling work has developed foundational identifiability conditions and algorithms, robustness remains a key barrier in real deployments. Most analyses treat *additive* measurement noise (often stochastic) and study the stability of operator recovery under such perturbations. For example, [21] analyzes the impact of additive random noise in the finite-dimensional setting, including recovery of driving operators defined via real symmetric convolution, and integrates denoising strategies to mitigate noise effects. However, many sensing systems also experience *gross, intermittent corruptions* that are better modeled as *sparse outliers in time*: a sensor may malfunction for a small number of time instants, packets may be dropped or replaced by erroneous values, or a transient interference event may corrupt a few snapshots. These sparse but potentially large-magnitude corruptions do not fit well into the existing dynamical sampling models and can catastrophically bias spectral/Prony-type identification. To address the challenge posed by the outliers, this paper focuses on the following setting: Measurements are collected at fixed

subsampled spatial locations across time, but a small subset of time snapshots are corrupted by outliers, no matter whether one or more sensors are impacted at the time.

Later in Section IV, we show that the spectral recovery against time-sparse outliers can be lifted as a series of robust low-rank Hankel matrix recovery problems, with the ranks explicitly determined by spectral collisions induced by aliasing. Fortunately, robust low-rank Hankel recovery has been extensively exploited in various methodologies, such as nuclear norm minimization [22], gradient descent [23], [24], Newton-like method [25], Riemannian optimization [26], and alternating projection [27], [28].

Contributions. Our main contributions are threefold:

- We propose a novel robust spectral recovery model for dynamical sampling with time-sparse corruptions, which induce a sequence of robust low-rank Hankel recovery problems, whose low-rankness is guaranteed by Theorem 1.
- We propose a novel pipeline that uses a sequence of robust Hankel recovery and completion algorithms to enable stable Prony-style spectral recovery, with a provable upper bound on the fraction of tolerable outliers, grounded by Theorem 3.
- Numerical results under various settings show that the proposed approach tracks the true spectrum of the underlying convolution operator more accurately and robustly against the state-of-the-art, even with outliers and additive noise.

II. NOTATION AND PRELIMINARIES

Let $d = mJ$ with integers $m \geq 2$ and $J \geq 1$. We work on cyclic groups $\mathbb{Z}_d$ and $\mathbb{Z}_J$ (indices modulo d and J). We use $\|\cdot\|_2$, $\|\cdot\|_F$, and $\langle A, B \rangle = \text{trace}(A^*B)$.

a) *Circular convolution and Fourier transform.*: For $a, f \in \ell^2(\mathbb{Z}_d)$, define circular convolution

$$(a * f)(k) = \sum_{t \in \mathbb{Z}_d} a(t)f(k-t).$$

Let $\widehat{(\cdot)}$ denote the DFT on $\mathbb{Z}_d$. Then $\widehat{a * f}(k) = \widehat{a}(k)\widehat{f}(k)$, so the convolution operator $Af := a * f$ is diagonalized by the DFT with spectrum $\{\widehat{a}(k)\}_{k \in \mathbb{Z}_d}$ [1], [20].

b) *Uniform subsampling.*: The uniform subsampling operator $S_m : \ell^2(\mathbb{Z}_d) \rightarrow \ell^2(\mathbb{Z}_J)$ is $(S_m z)(j) = z(mj)$, $j \in \mathbb{Z}_J$, and satisfies the identity

$$\widehat{(S_m z)}(j) = \frac{1}{m} \sum_{n=0}^{m-1} \widehat{z}(j + nJ), \quad j \in \mathbb{Z}_J, \quad (1)$$

where $\widehat{z}$ on the right is the DFT on $\mathbb{Z}_d$ [1], [20].

c) *Hankel lifting.*: Given a scalar sequence $\{s_\ell\}_{\ell=0}^{L-1}$ and K with $2K \leq L$, define the $K \times K$ Hankel matrix

$$H_K(s) := [s_{p+q}]_{p,q=0}^{K-1}. \quad (2)$$

We write $\mathbb{H}_K$ for the space of all $K \times K$ Hankel matrices.

III. PROBLEM SETUP

A. Dynamical sampling with time-sparse corruptions

Let A be the convolution operator $Af = a * f$ on $\ell^2(\mathbb{Z}_d)$. Given an unknown initial state $f \in \ell^2(\mathbb{Z}_d)$, the orbit is

$$x_\ell = A^\ell f, \quad \ell \in \{0, \dots, L-1\} \text{ with } [L] := \{1, \dots, L\}.$$

We observe uniform subsamples

$$y_\ell = S_m x_\ell \in \ell^2(\mathbb{Z}_J), \quad \ell \in \{0, \dots, L-1\}, \quad (3)$$

where S_m is defined in Section II. We focus on *time-sparse outliers*: there exists an unknown set $\Omega_{\text{out}} \subset \{0, \dots, L-1\}$ with $|\Omega_{\text{out}}| \ll L$ such that

$$\tilde{y}_\ell = y_\ell + e_\ell, \quad e_\ell = 0, \quad \text{for } \ell \notin \Omega_{\text{out}}, \quad (4)$$

and the nonzero e_ℓ can have arbitrary magnitudes (gross snapshot corruptions).

B. Low-rank Hankel structure in Fourier-domain

Define $s_\ell(j) := \widehat{y}_\ell(j)$, $\tilde{s}_\ell(j) := \widehat{\tilde{y}}_\ell(j)$, $j \in \mathbb{Z}_J$. By (1) and $\widehat{A^\ell f}(k) = \widehat{a}(k)^\ell \widehat{f}(k)$, each $\{s_\ell(j)\}_{\ell \geq 0}$ is a finite mixture of exponentials in ℓ , hence the Hankel matrix $H_K(j) := \mathcal{H}_K(\{s_\ell(j)\})$ is low rank (Theorem 1). Moreover, (4) implies

$$\tilde{H}_K(j) := \mathcal{H}_K(\{\tilde{s}_\ell(j)\}) = H_K(j) + O_K(j),$$

where $O_K(j) \in \mathbb{H}_K$ has nonzero anti-diagonals only on Ω_{out} .

C. Goals

Given the corrupted samples $\{\tilde{y}_\ell\}_{\ell=0}^{L-1}$, our goals are:

- **Robust Hankel recovery:** recover the low-rank Hankel matrices $\{H_K(j)\}_{j \in \mathbb{Z}_J}$ and identify the outlier set Ω_{out} from the structured observations $\{\tilde{H}_K(j)\}$.
- **Robust spectral recovery:** estimate the aliased spectra $\{\widehat{a}(j+nJ)\}_{n=0}^{m-1}$ for each j , and recover the spectrum (and, when identifiable, the filter a) of the driving operator A .

IV. METHODOLOGY

This section addresses the two goals in Section III: (i) robust recovery of the clean Hankel matrices and the outlier time indices, and (ii) recovery of the spectrum of the driving convolution operator A .

A. Low-rank Hankel structure

We first recall the key structural fact: after uniform subsampling, each Fourier channel produces a mixture of exponentials in time, which implies low-rank Hankel structure.

Theorem 1 (Low-rank dynamical Hankel matrices). *Let $d = mJ$ with d odd, and let $a, f \in \ell^2(\mathbb{Z}_d)$. Fix $j \in \mathbb{Z}_J$ and define*

$$y_\ell = S_m(a^\ell * f), \quad s_\ell(j) := \widehat{y}_\ell(j), \quad \ell \in \{0, \dots, L-1\}.$$

For an integer $K \geq 1$ with $2K \leq L$, define the Hankel matrix $H_K(j) := [s_{p+q}(j)]_{p,q=0}^{K-1}$. Let r_j be the number of distinct values among $\{\widehat{a}(j+nJ)\}_{n=0}^{m-1}$.

Then for any $K \geq r_j$, $\text{rank}(H_K(j)) \leq r_j$, and for generic f equality holds. In particular, if $\hat{a}$ is real symmetric and strictly decreasing on $[0, \frac{d-1}{2}]$, then

$$r_j = \begin{cases} \frac{m+1}{2}, & j = 0; \\ m, & j > 0. \end{cases}$$

Theorem 1 yields a *rank separation* between $j = 0$ and $j > 0$ under mild spectral monotonicity, which we exploit for robust outlier localization.

B. Robust Hankel recovery with time-sparse outliers

Recall the time-sparse corruption model (4). After DFT on $\mathbb{Z}_J$ and Hankel lifting, for each $j \in \mathbb{Z}_J$,

$$\tilde{H}_K(j) = H_K(j) + O_K(j),$$

where $O_K(j) \in \mathbb{H}_K$ is supported only on anti-diagonals indexed by the outlier set Ω_{out} (equivalently, few corrupted time indices).

Let $\mathcal{I} := \{0\} \cup [2K-2]$ be the anti-diagonal index set. We first detect corrupted anti-diagonals using the channel $j = 0$, and then complete the remaining channels using the detected clean indices.

Stage I: Reference Hankel matrix at $j = 0$. Under the assumptions of Theorem 1, the Hankel matrix $H_K(0)$ has the smallest rank $\frac{m+1}{2}$, making it well suited for outlier detection, since lower rank Hankel matrices can tolerate a larger fraction of sparse outliers [22]. Accordingly, the outlier detection task can be formulated as the following optimization problem:

$$\begin{aligned} \min_{H, O \in \mathbb{H}_K} \quad & \frac{1}{2} \|\tilde{H}_K(0) - H - O\|_F^2 \\ \text{s.t.} \quad & \text{rank}(H) = \frac{m+1}{2}, O \text{ is sparse.} \end{aligned} \quad (5)$$

To solve (5), we employ the *Accelerated Structured Alternating Projections (ASAP)* algorithm proposed in [28], which is specifically designed for outlier detection of low-rank Hankel matrices. The recovered sparse component $\hat{O}$ identifies the corrupted anti-diagonals of the Hankel matrix and thereby provides an estimate $\hat{\Omega}_{\text{out}}$ of the outlier support.

Stage II: Hankel matrix completion for $j > 0$. Once the outlier support has been estimated from the reference frequency $j = 0$, we turn to the recovery of the remaining Hankel matrices. For $j > 0$, we remove the detected outlier indices and solve a low rank Hankel completion problem:

$$\begin{aligned} \min_{H \in \mathbb{H}_K} \quad & \frac{1}{p} \langle \mathcal{P}_{\Omega^c}(H - \tilde{H}_K(j)), H - \tilde{H}_K(j) \rangle \\ \text{s.t.} \quad & \text{rank}(H) = m, \end{aligned} \quad (6)$$

where Ω corresponds to anti-diagonals indexed by $\hat{\Omega}_{\text{out}}$, and $p := \frac{\Omega^c}{L}$ is the rate of uncorrupted observation. The remaining task is therefore to solve (6). To this end, we employ the *Hankel Structured Newton-Like Descent (HSNLD)* algorithm proposed in [25], which is designed for robust low-rank Hankel matrix completion.

C. Spectrum recovery from cleaned Hankel matrices

We now describe how to recover the spectrum of the driving operator A once the time-domain outliers have been removed via robust Hankel recovery. Throughout this subsection we focus on the convolution model $Ax = a * x$ with real-symmetric filter a , so that A is diagonalized by the DFT and $\sigma(A) = \{\hat{a}(k)\}_{k \in \mathbb{Z}_d}$. According to previous analysis, each fixed frequency $j \in \mathbb{Z}_J$ yields a scalar time sequence $\{s_\ell(j)\}_{\ell \geq 0}$ that is a linear combination of exponentials in ℓ , i.e., $s_\ell(j) = \frac{1}{m} \sum_{n=0}^{m-1} \hat{a}(j+nJ)^\ell \hat{f}(j+nJ)$. Consequently, $\{s_\ell(j)\}$ satisfies a short linear recurrence whose characteristic polynomial has roots equal to the *distinct* values among $\{\hat{a}(j+nJ)\}_{n=0}^{m-1}$ (cf. [19], [21]).

Step 0: Input from Hankel recovery. From Section IV-B, we obtain denoised sequences $\{\hat{s}_\ell(j)\}_{\ell=0}^{L-1}$ for each $j \in \mathbb{Z}_J$ (or equivalently denoised Hankel matrices $\hat{H}_K(j)$). In what follows we write $s_\ell(j)$ for the cleaned/denoised sequence.

Step 1: Estimate the model order r_j . Let $r_j := \text{rank}(H_K(j))$ (in the noiseless case, r_j equals the number of distinct values among $\{\hat{a}(j+nJ)\}_{n=0}^{m-1}$; see Theorem 1). In practice, we estimate r_j by a numerical rank test on $\hat{H}_K(j)$.

Step 2: Compute the annihilating polynomial. For each j , define the coefficient vector $c(j) = (c_0(j), \dots, c_{r_j-1}(j))^\top$ by fitting the recurrence

$$s_{k+r_j}(j) + \sum_{\ell=0}^{r_j-1} c_\ell(j) s_{k+\ell}(j) = 0, k \in \{0\} \cup [K - r_j - 1],$$

which is the finite-dimensional analogue of the minimal-polynomial relation used in [21]. This can be written as a Hankel linear system

$$[H_K(j)]_{:,1:r_j} c(j) = [H_K(j)]_{:,r_j+1}. \quad (7)$$

If $K > 2r_j$, we solve (7) in the least-squares sense for robustness to residual denoising error.

We then form the monic polynomial

$$p_j(\lambda) = \lambda^{r_j} + \sum_{\ell=0}^{r_j-1} c_\ell(j) \lambda^\ell.$$

Step 3: Recover the spectral values. Let $R(j)$ be the set of roots of p_j . In the noiseless case,

$$R(j) = \{\lambda_{j,t}\}_{t=1}^{r_j} = \cup_{n=0}^{m-1} \{\hat{a}(j+nJ)\}. \quad (8)$$

After robust Hankel recovery, $R(j)$ provides a stable estimate of these values.

D. Recovery guarantees of low rank Hankel matrix

To ensure a low-rank matrix is recoverable from partial observations, we first recall the incoherence condition, which is widely used in matrix completion [29]. For Hankel matrices,

this condition can be defined in terms of Vandermonde decomposition. Note the Hankel entries satisfy $H_K(j)_{p,q} = s_{p+q}(j)$ with

$$s_{p+q}(j) = \frac{1}{m} \sum_{n=0}^{m-1} \hat{a}(j+nJ)^{p+q} \hat{f}(j+nJ) = \sum_{t=1}^{r_j} \alpha_{j,t} \lambda_{j,t}^{p+q}.$$

Therefore,

$$H_K(j) = V_K(j)D(j)V_K(j)^\top$$

with $V_K(j) = [\lambda_{j,t}^p]_{p=0, t=1}^{K-1, r_j}$ and $D(j) = \text{diag}(\alpha_{j,1}, \dots, \alpha_{j,r_j})$. Then, the incoherence condition for the underlying Hankel matrix $H_K(j)$ is defined as follows.

Definition 2 (Incoherence [22], [23]). *The Hankel matrix $H_K(j)$ with the Vandermonde decomposition $H_K(j) = V_K(j)D(j)V_K(j)^\top$ is said to be μ -incoherent if the smallest singular values $\sigma_{\min}(V_K^\top V_K) \geq \frac{K}{\mu}$.*

Let κ_j be the condition number of $H_K(j)$, $j \in \mathbb{Z}_J$, denote $\kappa := \max_j \{\kappa_j\}$. Then, we have the recovery guarantee.

Theorem 3 (Recovery guarantee for Hankel matrices). *Suppose $\{H_K(j)\}_{j \in \mathbb{Z}_J}$ are μ -incoherent, and it holds*

$$|\Omega_{\text{out}}| \leq \min \left\{ L - \mathcal{O}(\mu^2 m^2 \kappa^4 \log L \log \frac{1}{\varepsilon}), \right. \\ \left. \mathcal{O}\left(\frac{4L}{\mu^2(m+1)^2 \kappa^2}\right) \right\}$$

with $L \gg \log(1/\varepsilon)$. Then, with probability at least $1 - \frac{J}{L}$, the solutions to (5) and (6) obtained via ASAP and HSNLD yield ε -accurate estimates $\{\hat{H}_K(j)\}_{j \in \mathbb{Z}_J}$ of $\{H_K(j)\}_{j \in \mathbb{Z}_J}$, that is,

$$\|\hat{H}_K(j) - H_K(j)\|_F \leq \varepsilon, \quad \forall j \in \mathbb{Z}_J.$$

V. NUMERICAL EVALUATION

We evaluate the proposed Hankel-Prony pipeline for spectral recovery under *time-sparse outliers* and *additive Gaussian noise*. Throughout, the convolution operator is diagonalized by the DFT, and its spectrum is denoted by $\hat{a} \in \mathbb{R}^n$ (real-symmetric in our construction). We report results in terms of (i) spectrum plots and (ii) spectral error/SNR metrics.

A. Data generation and corruption model

We work on a cyclic grid $\mathbb{Z}_d$ with d odd and generate a real-symmetric spectrum $\hat{a}$ by drawing $\hat{a}(0) = 1$, sampling i.i.d. entries on $\{1, \dots, (d-1)/2\}$, and mirroring them to enforce $\hat{a}(k) = \hat{a}(d-k)$. We simulate convolutional dynamics as following:

$$x_{\ell+1} = \mathcal{F}^{-1}(\hat{a} \odot \mathcal{F}x_\ell), \quad \ell = 0, 1, \dots, L-1,$$

with $L = 300$ starting from a random x_0 . Uniform spatial subsampling uses $J = d/m$ sensors at locations $\{0, m, 2m, \dots, (J-1)m\}$, producing $y_\ell = S_m x_\ell \in \mathbb{R}^J$. We then inject:

- **Time-sparse outliers:** Choose an outlier index set $\Omega_{\text{out}} \subset \{0, \dots, L-1\}$ uniformly at random with $|\Omega_{\text{out}}| = \lfloor \alpha L \rfloor$. For each $\ell \in \Omega_{\text{out}}$, we corrupt the snapshot by adding a perturbation $e_\ell \in \mathbb{R}^J$ whose entries are drawn i.i.d. from a

TABLE I: Signal-to-noise ratios (dB) for the measurements and the recovered spectra under mixed sparse outliers and Gaussian noise. As σ decreases, the proposed method continues to improve and tracks the true spectrum increasingly accurately, whereas the Cadzow baseline exhibits limited gains due to its sensitivity to sparse, structured outliers.

σ	SNR _{Gauss}	SNR _{Outlier}	SNR _{Proposed}	SNR _{Cadzow}
10^{-3}	42.8783	7.7888	19.4622	4.8494
10^{-5}	82.7459	7.7888	56.6587	4.8893
10^{-7}	123.2371	7.7888	90.4528	4.8886
10^{-9}	162.6056	7.7888	128.4645	4.8886

uniform distribution with scale proportional to the snapshot magnitude: $(e_\ell)_j \sim \text{Unif}(-c\mu_\ell, c\mu_\ell)$, $j \in [J]$, where $\mu_\ell := \frac{1}{J} \|y_\ell\|_1$. For $\ell \notin \Omega_{\text{out}}$, we set $e_\ell = 0$.

- **Additive Gaussian noise:** Add i.i.d. Gaussian noise g_ℓ with standard deviation $\sigma \in \{10^{-3}, 10^{-5}, 10^{-7}, 10^{-9}\}$.

The observed samples are $\tilde{y}_\ell = y_\ell + e_\ell + g_\ell$.

B. Methods compared

- 1) **Proposed (robust Hankel + completion + Prony).** We apply robust Hankel denoising on the reference frequency channel (the lowest-rank Hankel case), estimate the corrupted time set, complete the remaining channels using Hankel completion, and finally recover the spectral values using the annihilating-filter/Prony step.
- 2) **Cadzow baseline (Cadzow + Prony).** We use the state-of-the-art *Cadzow denoising* method (alternating projections onto the low-rank and Hankel constraints) as our baseline, which is applied independently to each channel and followed by the same Prony step. This route is a standard and widely used stabilization heuristic for Prony/Hankel-based spectral estimation and has also appeared in the dynamical sampling literature as a denoising pre-step (often referred to as *Cadzow-like denoising*); see [20], [21], [30], [31].

C. Evaluation metrics

Let $s = \{\hat{a}(k)\}_{k=0}^{d-1}$ and $\hat{s}$ be the corresponding estimate. We will report:

- **Relative spectral error:** $\text{RE} := \|\hat{s} - s\|_2 / \|s\|_2$.
- **Spectral SNR (dB):** $\text{SNR}_{\text{spec}} := -20 \log_{10} \text{RE}$.

To quantify measurement corruption strength in the observation domain, let $Y = [y_0, \dots, y_{L-1}]$, $E = [e_0, \dots, e_{L-1}]$, $G = [g_0, \dots, g_{L-1}]$, and $\tilde{Y} = Y + E + G$. We will report $\text{SNR}_{\text{Outlier}} := 20 \log_{10}(\|Y\|_F / \|E\|_F)$ and $\text{SNR}_{\text{Gauss}} := 20 \log_{10}(\|Y\|_F / \|G\|_F)$.

D. Spectrum recovery under mixed corruptions

We fix $(d, m, J, L) = (15, 3, 5, 300)$ and a time-outlier rate $\alpha = 5\%$, and test several Gaussian noise levels. For each σ , we plot the recovered spectrum against the ground truth and record the corresponding SNRs.

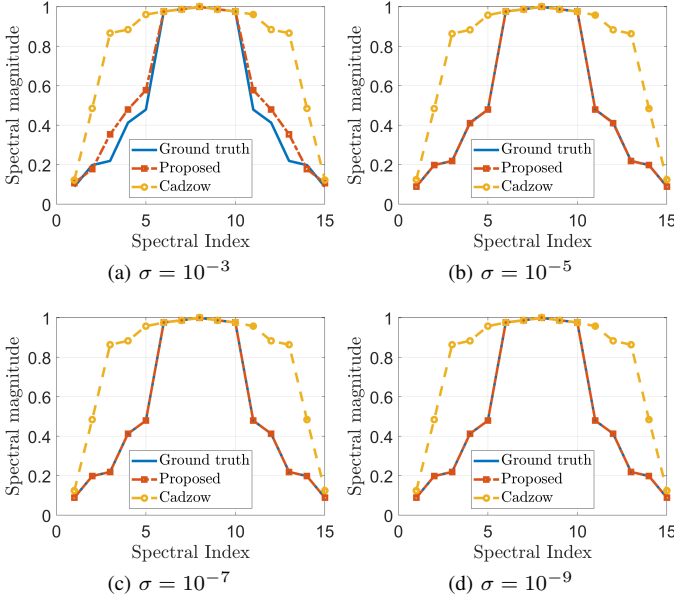


Fig. 1: Spectral recovery under time-sparse outliers (rate $\alpha = 5\%$) and additive Gaussian noise at three noise levels. Each panel plots the spectral magnitude $\hat{a}(k)$ (ground truth) and the recovered spectra from **Proposed** (robust Hankel denoising + Prony) and **Cadzow** (Cadzow denoising + Prony). As the Gaussian noise level decreases, the proposed method increasingly matches the true spectrum and resolves the correct spectral peaks, whereas the Cadzow baseline remains biased due to its sensitivity to sparse, structured corruptions.

Figure 1 shows representative recovered spectra at three noise levels ($\sigma = 10^{-3}, 10^{-5}, 10^{-7}, 10^{-9}$). Across all cases, our method closely tracks the ground truth despite time-sparse outliers, whereas the Cadzow baseline is noticeably biased under sparse, potentially large corruptions. The SNR summaries in Table I corroborate this trend: as σ decreases (i.e., the Gaussian noise weakens and the measurement SNR increases), the recovered-spectrum SNR of our method improves substantially, while the Cadzow-based method remains much lower.

E. Robustness versus corruption rate

We fix $(d, m, J, L) = (21, 3, 7, 300)$ and sweep the corruption rate $\alpha \in \{0.01, 0.03, \dots, 0.15\}$. For each α , we run 15 independent trials with *sparse-in-time snapshot outliers only* and compute the spectral recovery relative error RE. We summarize variability across trials using a semilogy plot of the *median* RE versus α ; see Figure 2. This experiment isolates the effect of increasing time-sparse corruptions and shows that explicitly modeling the induced Hankel outliers yields substantially improved robustness compared with purely Cadzow denoising-based baselines, for both mild outliers ($c = 1$, **Left**) and more severe corruptions ($c = 5$, **Right**).

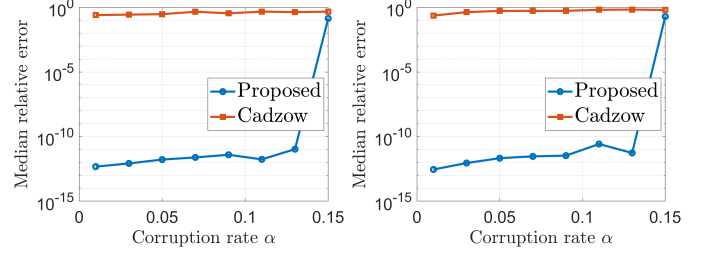


Fig. 2: Median relative error of spectral recovery versus corruption rate α , computed over 15 independent trials per α , with outlier amplification factors $c = 1$ (**Left**) and $c = 5$ (**Right**). Across a range of α , the proposed method maintains near machine-precision accuracy, whereas the Cadzow baseline exhibits order-one error and is largely insensitive to decreasing α , reflecting its fragility to sparse snapshot corruptions.

VI. CONCLUSION AND FUTURE WORK

This paper proposes a novel robust spectral recovery model against time-sparse corruptions for dynamical sampling on a finite cyclic grid. A robust pipeline has been developed to solve the spectral recovery problem. The pipeline employs a sequence of robust Hankel recovery and completion algorithms to enable the stable Prony-style spectral recovery. A provable upper bound on the fraction of tolerable outliers has been established. The performance of the proposed methodology has been verified with empirical experiments. In particular, the proposed method tracks the true spectrum of the underlying convolution operator accurately and robustly, and outperforms the state-of-the-art constantly.

Future work will include a detailed theoretical analysis to bridge the spectrum of the convolution operator and the parameters of the lifted Hankel matrices (such as μ and κ). The analysis results will be used to motivate an improved design for the robust pipeline and algorithms. Extended empirical experiments, including real-world applications, will be evaluated. We will also extend the settings to high-dimensional dynamical sampling problems.

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